

Question

It is known that **M** and **N** are adjacent elements in the periodic table with increasing atomic numbers, and they can form elemental gases **A** and **B**, respectively. At 25°C, heating a mixture of gold, elemental gas **A**, and elemental gas **B** yields a 1:1 type salt **X**. The cation of **X** is a dinuclear cation exhibiting paramagnetism, while the anion is a diamagnetic octahedral ion. When **X** is heated under vacuum conditions, it decomposes into **A**, **B**, and **C**, where **C** is a dinuclear complex in which the oxidation state of gold is the same as in **X**.

Based on the above information, select the correct option.

- A.** The bond order of the **X** cation is 1.5
- B.** The sum of the atomic numbers of **M** and **N** is 19.
- C.** According to the crystal field theory, the electron configuration of the central atom in the **X** anion is $t_{2g}^4 e_g^2$.
- D.** The molecular point group of the **C** molecule is D_{2d}
- E.** The relative molecular mass of **X** molecule is 343g/mol
- F.** In the reaction equation for the high-temperature decomposition of **X**, assuming the equation coefficients have been reduced to their simplest form, the sum of the coefficients of reactants and products is 5.
- G.** None of the above options is correct

Answer

Correct Answer: **E**

Detailed Explanation

According to the problem, there are not many pairs of adjacent elements in the periodic table that can form elemental gases. Considering the chemical properties of gold, it can be inferred that these two gases are highly oxidizing. It can be speculated that **N** is the element F , **A** is O_2 , and **B** is F_2 . The sum of the atomic numbers of these two elements is 17, making option B incorrect.

CHECKPOINT

1 PTS

A is O_2 , **B** is F_2

CHECKPOINT

1 PTS

The sum of the atomic numbers of the two elements is 17

The product **X** formed by the reaction of gold with gases **A** and **B** is an ionic compound. The anion is an octahedral complex, with the central element of the anion being Au surrounded by six ligands. The ligands cannot be oxygen, as this would result in an excessively high oxidation state for gold. Therefore, the composition of the anion should be AuF_6 , with the oxidation state unknown. As for the cation, it is dinuclear and should be O_2 , exhibiting paramagnetism. It carries a single positive charge, forming O_2^+ , having lost one antibonding electron, which increases the bond order by 0.5, resulting in a bond order of 2.5. Thus, option A is incorrect.

CHECKPOINT

1 PTS

The cation of **X** is O_2^+ , with a bond order of 2.5

The anion of **X** is AuF_6^- , where *Au* is in the +5 oxidation state. The chemical formula of **X** is O_2AuF_6 , with a relative molecular mass of 343 g/mol, making option E correct.

CHECKPOINT

1 PTS

The chemical formula of **X** is O_2AuF_6 , with a relative molecular mass of 343 g/mol

According to crystal field theory, the d orbitals of *Au* will split into three t_{2g} orbitals and two e_g orbitals in an octahedral field. Based on the oxidation state of *Au*, the valence electron configuration of the *Au* atom can be inferred as $(5d^6)$. If it is high-spin, the configuration would be $t_{2g}^4e_g^2$, with four unpaired electrons, exhibiting paramagnetism. If it is low-spin, the configuration would be t_{2g}^6 , with no unpaired electrons, making it diamagnetic. Given the diamagnetic condition, it can be inferred that the electrons of the *Au* element are in a low-spin state, with a hybridization of d^2sp^3 , making option C incorrect.

CHECKPOINT

1 PTS

The high-spin configuration is $t_{2g}^4e_g^2$, exhibiting paramagnetism

CHECKPOINT

1 PTS

The low-spin configuration is t_{2g}^6 , exhibiting diamagnetism

CHECKPOINT

1 PTS

The electron configuration of the **X** anion should be t_{2g}^6

When **X** is heated in a vacuum to decompose it, the oxidation state of *Au* in **C** is the same as in **X**, also +5. **C** should be a fluoride of *Au*, and since it is a dinuclear complex, it should be Au_2F_{10} . Two $[AuF_6]$ octahedra are connected by sharing an edge, and the molecular point group is D_{2h} , making option D incorrect.

CHECKPOINT

1 PTS

C is Au_2F_{10} , with a molecular point group of D_{2h}

The decomposition equation is written as $2O_2AuF_6 = Au_2F_{10} + 2O_2 + F_2$. The sum of the coefficients of the reactants and products is $2 + 1 + 2 + 1 = 6$, making option F incorrect.

CHECKPOINT

1 PTS

$2O_2AuF_6 = Au_2F_{10} + 2O_2 + F_2$, option F is incorrect